

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element E Initial Template - AT - 15 points

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

High-Priority Design Requirement #1 from Element C - The program must correctly guide users to over 90% of classrooms.

STEM Concepts Involved in the requirement:

- MATLAB software, school computers.
- System design, floor layout planning.
- Graph theory, coordinates.

Describe how STEM concepts apply directly to the specifics of your design requirement:
We would need to make the MATLAB software run accurately.

High-Priority Design Requirement #2 from Element C - The directions from the program must be clear.

STEM Concepts Involved in the requirement:

- MATLAB software, school computers.
- System design, floor layout planning.
- Graph theory, coordinates.

Describe how STEM concepts apply directly to the specifics of your design requirement:
We would need to design the system to be clear and concise.

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High-Priority Design Requirement #3 from Element C - The users must be able to choose the ending classroom.

STEM Concepts Involved in the requirement:

- MATLAB software, school computers.
- System design, floor layout planning.
- Graph theory, coordinates.

Describe how STEM concepts apply directly to the specifics of your design requirement:
We would need to make a correct floor layout of the school.

Proposed Solution to STEM Principles and Practices (E2)

STEM Principles for Design Solution #1

STEM Concept:

- MATLAB software, school computers.
- System design, floor layout planning.
- Graph theory, coordinates.

Describe how STEM concept applies directly to the specifics of the first solution to your problem:

Combining all of these concepts together will allow us to create the best program possible for everyone.



You can only work on the next sections after you have built and tested your first prototype.
You need to skip to the “Evidence is Reviewed” section.

STEM Principles for Design Solution #2



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STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your solution:

STEM Principles for Design Solution #3 (if completed)

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your solution:



Evidence Design is Reviewed (E3)

Documentation #1 (Only one documentation section needed for full credit):

What format: Interview Notes

- She said that they are very good connections to make.
- She said that they are good at proving the solution.



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Experts with Credentials (E4)

Expert #1 (from documentation 1): Mrs. Weelick, she is an engineering teacher in the school.

Verification or Corrective Response to Design Review (E5)

Interview with Expert #1 informs corrective response: "Mrs. Weelick told us to add stairs into the instructions, which informed our decision to reconsider adding stairs into the program.

	Criteria	5	4	3	2	1	0
	3. Evidence that Design is Reviewed. There is _____ evidence that the application of STEM principles & practices (or suitable alternative) has been reviewed by expert(s).	<i>substantial</i>	<i>adequate</i>	<i>some</i>	<i>minimal</i>	<i>no - but review was requested -</i>	<i>no</i>
	4. Design Reviewed by Experts. _____ expert(s) have/has reviewed the design for application of STEM principles & practices.	<i>Two or more, with at least two with documented, relevant credentials</i>	<i>Two or more, with at least one with documented, relevant credentials</i>	<i>At least one with documented, clearly relevant credentials</i>	<i>At least one with documented credentials</i>	<i>Only one with unclear credentials</i>	<i>No</i>



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	5.Design Review Results in Verification. The design reviews provided _____.	<i>clear and detailed confirmation (verification) or substantial detail necessary to inform a corrective response</i>	<i>clear confirmation (verification) or detail necessary to inform a corrective response</i>	<i>partial confirmation (verification) or some detail necessary to inform a corrective response</i>	<i>does not provide clear confirmation (verification) or suggestions for a corrective response, but at least some response is provided</i>	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert, but includes notes of attempts to gain feedback</i>	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert</i>
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